

c language

Control structure

Repetition Control structure

- C language statements that allows a programmer to execute specific blocks of code a number of times
- Types :
 - **While**
 - **Do-While**
 - **For**

While structure

- **While loop**

- is a statement or block of statements that is repeated as long as some condition is satisfied
- general format :

```
while (boolean_expression)  
{  
    statement1;  
    statement2  
    ...  
}
```

While structure

- Examples:

```
int counter = 1;
while (counter <= 10) {
    printf("counter = " + counter);
    counter += 1;
}
```

```
int counter = 1;
while (counter*counter < 1000) {
    printf(counter * counter);
    counter++;
}
```

Do While structure

- **do-while loop**

- is similar to the while-loop
- have statements that are executed several times as long as the condition is satisfied
- general format:

```
do
{
    statement1;
    statement2
    ...
} while (boolean_expression);
```

Do While structure

- Example

```
int x = 0;

do {
    printf("x = " + x);
    x++;
} while(x < 10);
```

```
do {
    printf("Hello!");
} while (false);
```

For structure

- **for statement**

- instructs a program to perform a block of code a specified number of times
- general format:

```
for (initialization; condition; increment)  
{  
    statement 1;  
    statement 2;  
    ...  
}
```

For structure

- Examples:

```
int x;  
for(x=1; x<=10; x++)  
{  
    printf("x= " + x);  
}
```

```
int i;  
for(i=0; i<10; i++)  
{  
    printf(i);  
}
```